EE5203 L08 Worksheet

PWM, output compare, and input capture - Basri Erdogan

Expected time: 30-45 minutes **Policy:** Attempt every question before consulting the worked solutions.

Timer clock = **16 MHz** throughout.

Part A - PWM arithmetic

1. Write the two formulas: PWM frequency in terms of `PSC` and `ARR`, and duty cycle in terms of `CCR` and `ARR`.
2. Complete the table. Show the arithmetic.

Requirement	`PSC`	`ARR`	`CCR`
1 kHz, 35 %			
2 kHz, 60 %			
200 Hz, 25 %			
50 Hz servo, 1.5 ms pulse			

3. Explain why `CCR = ARR` does not give 100 % duty. What value does it give for `ARR = 999`, and what would give a true 100 %?
4. A student wants to change the PWM *frequency* at run time and writes a new value into `CCR`. Describe what actually changes, and name the register they should have written.
5. TIM3 has four channels. A team wants a 1 kHz LED on CH1 and a 50 Hz servo on CH2, both on TIM3. Explain why this cannot work, and give the fix.
6. At `PSC = 15`, `ARR = 19999`, explain in one sentence why `CCR` can be read directly as "the pulse width in microseconds."

Part B - The handover

7. Name the register and bit-field that show a pin has been handed from GPIO output to a timer, and give the value it takes.
8. A student's PWM does nothing, but `HAL_GPIO_WritePin` still works on the same pin. Give the diagnosis and the one thing to check in the `.ioc`.
9. `HAL_TIM_PWM_Start()` is called but `CCER` bit 0 reads 0 afterwards. What would you suspect, and what is the visible symptom?

Part C - Capture mechanism

10. In one sentence each: what does `CC1S` in `CCMR1` select, and what does `CC1P` in `CCER` select in capture mode?
11. Who writes `CCR1` in capture mode, and at exactly what instant?
12. Explain **direct** and **indirect** capture mode. If CH1 is direct on PB6, what does CH2 listen to when configured indirect — and what would it listen to if configured direct instead?
13. Why are two channels needed to measure duty cycle, when one is enough to measure period?

Part D - Capture arithmetic

14. A 16-bit timer runs at a 1 µs tick. Rising captures arrive at `4000` and `9000`; the falling capture between them is `5250`. Compute the period, frequency, high time, and duty cycle.
15. Rising captures arrive at `64000` then `1500` on the same timer.
    a. Compute the period with the `uint16_t` cast.
    b. Compute what a plain `int` subtraction would give.
    c. Explain the difference in terms of modular arithmetic.
16. What is the longest interval a 16-bit timer at a 1 µs tick can measure honestly? Give the answer in milliseconds and as a signal frequency.
17. A team measures a 10 Hz signal with that setup and gets a stable, plausible reading of about 35 ms. Explain what went wrong and give **two** different fixes.

18. Your capture code works on TIM4 (16-bit). The team moves it to TIM2 (32-bit) unchanged, keeping the `(uint16_t)` casts. What now goes wrong, and when?

Part E - Evidence and behaviour

19. Fill in the evidence you would show for each claim.

Claim	SFR observation
The duty is the value I configured	
The pin belongs to the timer	
Hardware is timestamping without my code	
The capture edge is the one I selected	
A capture was lost	

20. Describe an experiment, using only the debugger and the SFR view, that makes `CC1OF` become set on purpose. Explain what was lost.
21. You flip `CC1P` for the capture channel while the program runs. Predict what happens to the measured **period**, and to the measured **duty**. Justify both answers.
22. The generating side sweeps its duty continuously (the breathing LED) while the measuring side reports duty. Explain why the reported duty tracks smoothly rather than jumping, and what would happen if the generator's `ARR` changed at the same time.

Exit self-assessment

- ☐ I can derive `PSC`/`ARR`/`CCR` for any frequency and duty.
- ☐ I can explain the 100 % off-by-one without looking it up.
- ☐ I can say who writes `CCR` in each direction, and when.
- ☐ I can compute period and duty from captures, wrap-safe.
- ☐ I can state the honest measurement range of a 16-bit capture timer.